

Same Budget, Smarter Spend

Making Malawi's Fertilizer Subsidy Work Harder

POLICY BRIEF · 2026 · Evidence from a randomized controlled trial (RCT) in 113 villages of central Malawi

The bottom line. Malawi does not need to spend more to get more from its fertilizer subsidy. In a randomized trial, a soil-test recommendation on its own changed nothing. The same recommendation paired with a flexible voucher worth the value of the Affordable Inputs Programme (AIP) bundle raised maize yields by about 19 percent, but farmers spent it on the same urea and nitrogen-phosphorus-potassium (NPK) products the AIP already distributes, not the site-specific blends their soil tests prescribed. Three implications follow: information is not the binding constraint; a flexible voucher can deliver the AIP's yield gains through private agro-dealers at comparable cost; and precision fertilization will not happen until the recommended products are on the shelf. A phased reform can cut waste and build private input markets **at no additional fiscal cost**.

Why this matters now

Fertilizer subsidies absorb a large share of Malawi's agricultural budget, and world prices remain volatile. Every kwacha spent subsidising the wrong nutrient, or fertilizer a household would have bought anyway, is a kwacha not spent elsewhere. With fiscal space tight, the question is no longer *whether* to subsidise fertilizer but *how to deliver the same support with less waste and more output*. We tested this directly: 113 villages were randomly assigned to (i) a plot-level soil-test recommendation, (ii) the same recommendation plus a voucher of 170,000 Malawi kwacha (MWK, about US\$100) redeemable for any fertilizer at a commercial agro-dealer, or (iii) a comparison group.

What we found

- **A soil test alone does nothing.** No change in fertilizer use, nutrients applied, maize yield, or subsidy participation. After two decades of a standard subsidised bundle, better information is not what is missing.
- **Money works, but farmers buy the familiar bundle.** The voucher raised fertilizer use by about 30 kg and maize yield by about 19 percent (roughly 151 kg per household). The extra fertilizer was almost entirely urea and NPK 23:10:5, a near-copy of the AIP bundle.
- **The precision products were prescribed but never bought.** Potassium was recommended for about two-thirds of households and lime for about a quarter; uptake of both, and of calcium ammonium nitrate (CAN) and monoammonium phosphate (MAP), was essentially zero. The barriers are practical: these products are rarely stocked, come in oversized packs, and are unfamiliar.
- **Much of the voucher displaced spending farmers would have made anyway.** This makes it very attractive to the household (farm profit rose about 190,000 MWK, half of it money saved on purchases no longer needed) but limits its *additional* contribution to national production.

The numbers: a cost-benefit snapshot

The programme's return to the public purse depends almost entirely on **how maize is valued**. Most beneficiaries are net maize buyers (about 44 percent of maize consumption is bought on the market), so the output they gain is worth far more at lean-season retail prices than at harvest-time farmgate prices.

Valuation scenario	Maize price (MWK/kg)	Value of +151 kg (MWK)	Voucher cost (MWK)	Benefit–cost ratio
Harvest-time farmgate (median)	750	113,000	170,000	0.67
Break-even	1,126	170,000	170,000	1.00
Lean-season retail (2025)	1,360	205,000	170,000	1.21
Peak retail (prior year)	1,742	263,000	170,000	1.55

Valued at the price farmers receive selling just after harvest, the voucher does *not* pay for itself; valued at the price the same households pay buying maize back in the lean season, it clears its cost comfortably. **The biggest lever on value is price and timing, not agronomy.** Two honest cautions: the soil test is an added cost that bought no behaviour change (it is excluded above and would lower every ratio), and the household's profit gain is partly a transfer of spending, not new production, so the reform case rests on delivering support *more efficiently*, not on the headline profit figure.

Five recommendations

1. **Do not pay for soil tests until the products they prescribe can be bought.** Recommendations changed no behaviour even with money in hand; the diagnostic is overhead until farmers can act on it. Sequence diagnostics *after* the supply chain.
2. **Fix the shelf before promoting precision.** Invest first in agro-dealer stocking of precision blends, plot-sized

packages, and on-farm demonstration. Precision fertilization is a supply-and-familiarity problem, not an information problem.

3. **Pilot flexible commercial vouchers as an AIP delivery channel, and measure leakage head-to-head.** At comparable cost the voucher matched the in-kind bundle's yield gains through the private sector; the case for switching is targeting, reduced diversion, and market development. Let a costed pilot decide the scale-up.
4. **Value and time the subsidy to the consumption calendar.** Returns more than double when output is valued at lean-season retail prices, which net-buyer households actually face. Pair input support with storage or timing that captures the high-value window.
5. **Score the programme on net profit and additional production, not kilograms distributed.** Track crowd-out-adjusted additionality so the budget is judged on *new* production it creates, not on inputs farmers would have bought anyway.

The opportunity

Malawi can make its largest agricultural expenditure work harder without spending a kwacha more, through a simple, politically safe sequence: **(1)** shift AIP delivery toward flexible commercial vouchers to capture targeting and market-building gains at equal cost (farmers still receive support, yields still rise); **(2)** reinvest the efficiency savings in the supply chain and farmer familiarity precision fertilization requires; **(3)** add soil-based diagnostics only once the recommended products can be bought. This turns a static, in-kind handout into a market-building, precision-ready system. It preserves what farmers and voters value about the subsidy and removes what the Treasury cannot afford: waste.

Methods and caveats. Findings are from a pre-registered cluster-randomized trial of roughly 2,000 households; effects are intention-to-treat and reflect one season in central Malawi. The voucher was delivered on top of the existing AIP, so the trial identifies the effect of adding a recommendation-paired voucher within the current regime; the “voucher as a full replacement for the AIP” comparison is a supported extrapolation, not a directly measured contrast. Price scenarios use observed sample and market prices. The soil-test unit cost is excluded from the ratios and should be added for any programme-level accounting.